ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

Faculty of Engineering

Department of Computer Science

MASTER'S PROJECT

Electricity Carbon Intensity Prediction for Germany Using AutoML

Emission Intensity Prediction Using AutoML

Supervisors:

Nick Harder
André Biedenkapp

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Submitted by:

Gurmeher Singh Puri
Udit Chawla

Abstract

This master's project develops a machine learning system for predicting electricity grid carbon emission intensities across Germany using automated machine learning techniques. The system addresses the critical challenge of forecasting carbon footprint variations in Germany's energy landscape, where renewable sources create significant daily and seasonal fluctuations in emission intensities.

The project integrates data from multiple European energy sources, including the ENTSO-E transparency platform and CO2map.de carbon tracking service, covering electricity generation forecasts, cross-border trading flows, and real-time carbon intensity measurements across Germany and eleven neighboring countries. The data pipeline processes information from over twenty-two cross-border electricity connections and four API endpoints to capture the complex interactions that influence German electricity carbon intensity.

The machine learning implementation employs AutoGluon's automated ensemble framework, configured with three gradient boosting algorithms (LightGBM, CatBoost, and XGBoost) to balance prediction accuracy with operational deployment requirements. The optimized configuration achieves model size reductions from several gigabytes to 100-130 MB per model while maintaining strong predictive performance suitable for real-time applications.

The system provides both production-based and consumption-based carbon intensity predictions at national and state levels, enabling applications in carbon-aware computing, energy management optimization, and environmental impact assessment. Feature engineering captures seasonal patterns, cross-border trading relationships, and renewable energy integration effects that drive carbon intensity variations throughout Germany's interconnected electricity system.

Results demonstrate the practical feasibility of accurate carbon intensity forecasting using publicly available energy market data, providing a foundation for carbon-aware decision making in modern energy systems. The configuration-driven architecture facilitates extension to additional European countries and supports future integration of advanced forecasting techniques and real-time deployment scenarios.

Acknowledgments

We acknowledge the CO2map.de platform and its development team for providing carbon intensity data and methodological frameworks that enabled this research. We thank ENTSO-E for maintaining the Transparency Platform that provides comprehensive European electricity market data.

We acknowledge the broader energy systems research community for developing the gap-filling algorithms and data processing methods that form the foundation of our data pipeline. We thank the developers of AutoGluon, LightGBM, CatBoost, and XGBoost for creating the machine learning frameworks that made this automated modeling approach possible.

Special thanks to our supervisors Nick Harder and André Biedenkapp for their guidance throughout this project. This work builds upon established methodologies in energy data analysis and automated machine learning, and we recognize the contributions of all researchers and practitioners who have advanced these fields.

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1 Introduction

The electricity sector plays a crucial role in global climate efforts, as it is responsible for approximately 25% of global greenhouse gas emissions while simultaneously being essential for economic development and quality of life. As countries transition from fossil fuels to renewable energy sources, the carbon intensity of electricity—measured in grams of CO2 equivalent per kilowatt-hour—varies dramatically throughout each day and across seasons.

Why Carbon Intensity Forecasting Matters: Accurate prediction of electricity carbon intensity enables several critical applications that support climate goals and efficient energy management. Carbon-aware computing systems can automatically shift computational workloads to periods when electricity is cleanest, reducing the environmental impact of data centers and cloud services. Energy storage systems can optimize charging and discharging schedules to maximize utilization of clean energy while minimizing reliance on fossil fuel generation. Smart grid operators can make informed decisions about demand response programs and grid balancing that prioritize low-carbon resources. Additionally, carbon intensity forecasts enable real-time carbon accounting for businesses and individuals, supporting informed decision-making about energy consumption timing.

This project develops a machine learning system that can predict how much carbon dioxide gets produced for each unit of electricity used in Germany. As Germany transitions from coal and gas to wind and solar power, the carbon footprint of electricity changes dramatically throughout each day. Sometimes electricity is very clean when the wind is blowing and the sun is shining. Other times, when renewable sources are not available, coal and gas plants must run, making electricity much more carbon-intensive.

The challenge is that Germany's electricity system does not work alone. It connects to eleven neighboring countries through high-capacity power lines. When Denmark has extra wind power, that clean electricity flows into Germany and reduces carbon emissions. When Germany needs more power and imports from Poland's coal plants, the carbon footprint goes up. The project captures all these complex relationships to make accurate predictions.

The system uses two main approaches. First, it collects data from official European energy platforms that track electricity generation, consumption, and trading across all European countries. Second, it uses AutoGluon, an automated machine learning framework that automatically tries many different prediction methods and combines the best ones into an ensemble model.

The project provides predictions for both the overall German market and each of the thirteen German states individually. Each region has different energy characteristics - northern states have lots of wind farms while southern states have more solar panels and industrial demand. The system accounts for these regional differences to provide accurate local predictions.

2 Data Sources and Collection Methods

Accurate carbon intensity prediction requires data that captures all the major factors affecting electricity generation and consumption across Europe. The project integrates information from multiple official sources, each providing different but essential pieces of the puzzle.

2.1 ENTSO-E Transparency Platform: European Energy Data

The European Network of Transmission System Operators for Electricity (ENTSO-E) operates a electricity data platform that coordinates electricity systems across forty-two transmission operators in thirty-five European countries. Unlike other commodity markets where products can be stored in warehouses, electricity must be perfectly balanced between generation and consumption every second of every day.

The project uses four main APIs from ENTSO-E to gather different types of electricity market information:

query_load_forecast: This API provides hourly predictions of how much electricity each country will need in the next day or two. These forecasts consider weather patterns, business activity, seasonal changes, and even major sports events that can cause millions of people to turn on their televisions at the same time. When demand is high, more power plants need to run, usually increasing carbon intensity.

query_generation_forecast: This API tells us how much total electricity each country expects to produce. It combines information from hundreds or thousands of individual power plants, each with different startup times and fuel requirements. Nuclear plants and coal plants take hours or days to start up, while gas plants can start faster. Renewable sources depend entirely on weather conditions.

query_wind_and_solar_forecast: This specialized API provides detailed predictions for wind and solar power output. Wind forecasts must translate weather predictions into electrical output estimates for wind turbines across different locations. Solar forecasts must account for cloud cover, seasonal sun angles, and different types of solar installations. These forecasts are crucial because renewable energy directly reduces carbon intensity.

query_day_ahead_prices: This API provides wholesale electricity market prices that show which power plants will actually run. European markets use a "merit order" system where the cheapest sources run first. Renewable sources with near-zero operating costs always run before expensive fossil fuel plants. Higher prices usually mean dirtier electricity because expensive plants are running.

The project also uses query_scheduled_exchanges to track electricity trading between Germany and eleven neighboring countries. This creates twenty-two different data streams because the system tracks both imports and exports for each connection. Each country has different energy characteristics that affect German carbon intensity when electricity flows across borders.

2.2 CO2map.de Carbon Intensity Integration and Attribution

While ENTSO-E provides detailed information about electricity generation and trading, it measures everything in megawatts of power. To predict carbon intensity, the project needs a way to convert those power measurements into environmental impact measured in grams of CO2 per kilowatt-hour.

Data Attribution: CO2map.de (https://www.co2map.de) provides this essential conversion by calculating both production-based and consumption-based carbon intensity. The platform was developed by researchers focused on electricity system decarbonization and implements methodologies described in Tranberg et al. (2019) and Schäfer et al. (2020). Production-based intensity measures the carbon footprint of electricity generated within German borders. Consumption-based intensity accounts for imports and exports, showing the true carbon footprint of electricity actually consumed by German users.

Technical Implementation: One technical challenge involves timing differences between data sources. CO2map.de carbon intensity data arrives with timestamps that are one hour behind the corresponding ENTSO-E forecast data. This happens because carbon intensity calculations need actual generation data, which becomes available after forecasts are made. The pipeline automatically adjusts these timestamps to ensure machine learning models learn the correct relationships between forecasts and the carbon intensity outcomes they predict.

Beyond national data, CO2map.de provides carbon intensity information for each of Germany's thirteen federal states. This detailed geographic information enables the development of specialized models for different regions with very different energy profiles.

2.3 Understanding Carbon Emission Intensities

Carbon emission intensity measures how much CO2 gets produced for each kilowatt-hour of electricity. This metric captures both direct emissions from burning fossil fuels and indirect emissions from building renewable energy infrastructure, processing nuclear fuel, and extracting and transporting fuels.

Carbon intensity follows predictable patterns throughout the day and across seasons. During sunny, windy periods, renewable sources provide most electricity needs, creating very low carbon intensity. During evening hours when people come home and turn on lights and appliances, demand peaks often require starting up fossil fuel plants, increasing carbon intensity. Winter periods typically show higher carbon intensity due to reduced solar generation and increased heating demand.

European electricity markets operate on economic principles where the cheapest generation sources run first. This creates a direct relationship between electricity demand and carbon intensity. During low demand periods, only clean and cheap sources like wind, solar, and nuclear operate. As demand increases, progressively more expensive and typically dirtier sources get activated, raising carbon intensity.

3 Technical Setup and Implementation

3.1 Development Environment and Software Requirements

The project uses Google Colab as the main development platform because it provides free access to powerful computers and integrates seamlessly with Google Drive for data storage. This cloud-based approach solves several practical problems including the need for substantial computing power during model training and reliable data storage that persists across work sessions.

The implementation requires several specialized Python libraries that provide essential functions for energy data analysis and machine learning. The entsoe-py library provides the main interface to the ENTSO-E platform, handling authentication, rate limiting, timezone conversions, and data formatting automatically. The autogluon.tabular module provides automated machine learning capabilities specifically designed for structured data like energy time series. It automatically tries many different model types and creates ensemble models that typically outperform individual algorithms. Additional libraries include pandas and numpy for data manipulation and numerical calculations, plus requests and sqlite3 for API communication and database storage.

The project organizes all files within a structured directory system on Google Drive. The main project folder contains three important subdirectories. The entsoe_DB_data_DE folder stores all processed datasets and CSV files. The autogluon_models folder contains trained machine learning models and related files. The project also maintains a central SQLite database that serves as the integration point where information from multiple sources gets combined.

3.2 Main Pipeline Workflow

The main pipeline operates as a complete system that manages data collection, processing, and model training through a single executable workflow. This approach ensures that results can be reproduced exactly and makes collaboration easier.

The pipeline begins by systematically collecting data from multiple APIs using smart retry mechanisms that handle rate limiting and temporary service outages. The system processes geographic relationships defined in configuration files to automatically generate appropriate API queries for all monitored countries and cross-border connections. For each neighboring country, the pipeline executes four different API calls covering load forecasts, generation forecasts, renewable energy forecasts, and electricity prices.

After collecting data, the pipeline builds a central SQLite database that serves as the integration point for all information. The database design handles the challenge of combining time series data from multiple sources with different update frequencies and geographic coverage areas. The master timeline construction represents one of the most complex parts of the entire pipeline. The process starts with a basic timeline and systematically adds information from all other data sources to create a dataset containing hundreds of features.

4 Data Processing and Feature Engineering

4.1 Time Management and Data Standardization

Working with energy data from eleven different European countries creates challenges around timing and time zones that must be resolved before any analysis can proceed. Different countries report data in their local time zones, daylight saving time changes happen on different dates, and some countries use different time standards for different types of data.

The solution involves converting all timestamps to Central European Time, which serves as the standard timezone for most continental European electricity markets. This standardization requires careful analysis of each dataset to determine its original timezone, then applying appropriate conversions while accounting for daylight saving time changes.

Various energy systems generate data at different intervals ranging from minutes to days. The solution creates consistent hourly time series while preserving important characteristics of the original data. For high-frequency data, the system calculates hourly averages but also keeps track of variation within each hour. For lower-frequency data, interpolation methods respect the physical constraints of power systems.

4.2 Handling Missing Data and Quality Control

Real-world energy data always has gaps due to equipment failures, communication outages, maintenance periods, or reporting delays. The approach to missing data management considers both the physical characteristics of energy systems and the requirements of machine learning algorithms.

Gap-Filling Implementation: The gap-filling algorithms used in this project is a simple yet effective pandas-based forward and backward fill. Short gaps in continuous variables like generation forecasts are handled through linear interpolation, which draws straight lines between known data points. This works well for gaps of a few hours because energy systems typically change gradually. Longer gaps require more methods that use historical patterns and seasonal relationships.

Beyond addressing missing data, quality control procedures identify and correct obviously incorrect values. This includes checks for physically impossible values like negative electricity generation or carbon intensities that exceed theoretical limits. The system implements automated error detection that flags data points requiring review.

4.2.1 Attribution for Data Processing Infrastructure

Gap-Filling Implementation: The gap-filling approach used in this project is based entirely on established methods from the broader time series and energy data analysis communities. We do not claim to have developed any novel gap-filling algorithm ourselves. Instead, we rely on widely used imputation techniques provided by the Pandas library, including forward-fill (ffill) and backward-fill (bfill), to address short, isolated gaps in energy intensity time series data.

In cases where more accurate estimation is required, we also apply linear interpolation methods, which are standard in time series forecasting and have been extensively described in the literature (e.g., Hyndman & Athanasopoulos, 2018). These approaches help maintain data continuity while avoiding the introduction of artificial trends.

All gap-filling methods used here are openly available in Python's data processing libraries and were selected based on their simplicity, reliability, and widespread adoption. The techniques and their foundations are fully credited to the original developers and researchers in this domain. Our implementation simply integrates these methods into the broader pipeline without modifying or extending them.

Timezone Normalization: The timezone conversion protocols follow established practices in multi-country energy analysis, converting all data to Central European Time as the standard for continental European electricity markets. These standardization methods represent common practice rather than novel contributions.

Data Quality Control: Beyond addressing missing data, quality control procedures identify and correct obviously incorrect values using methods established in energy data analysis literature. This includes checks for physically impossible values and automated error detection following standard practices in the field.

4.3 Creating Useful Features from Raw Energy Data

The feature engineering process transforms raw energy market data into input variables that capture the complex relationships affecting carbon intensity while providing machine learning algorithms with the information needed to make accurate predictions.

Complete Feature List and Engineering Contributions:

Temporal Features (Provided by existing infrastructure):

- Hour-of-day features (0-23) capturing daily demand cycles
- Day-of-week features (0-6) distinguishing weekday vs weekend patterns

- Seasonal features capturing annual cycles in renewable availability and demand

Cross-Border Flow Features (22 total):

- Import flows: DE_LU_imports_[country] for CH, BE, CZ, FR, NL, PL, DK_1, DK_2, AT, 10Y1001A1001A47J, 10YNO-2———T
- Export flows: DE_LU_exports_[country] for the same countries
- Aggregated features (engineered by our team): Total_Imports, Total_Exports

ENTSO-E Query Features (provided by APIs):

- Load forecasts for all countries (11 features)
- Generation forecasts for all countries (11 features)
- Wind and solar forecasts for all countries (33 features total: Solar, Wind_Onshore, Wind_Offshore where available)
- Day-ahead prices for all countries (11 features)

Carbon Intensity Features (targets):

- DE national: Generation_Intensity, Consumption_Intensity
- State-level: [State]_GenerationIntensity, [State]_ConsumptionIntensity for BB, BW, BY, HE, MV, NI, NW, RP, SH, SL, SN, ST, TH

Engineered Features (our contribution):

- Renewable penetration ratios: Solar_ratio, Wind_ratio calculated as renewable forecast / total generation forecast
- Net trading position: Net_position = Total_Imports - Total_Exports
- Price differentials: Price_spread = max(neighbor_prices) - min(neighbor_prices)
- Demand intensity: Load_per_capita calculated using population data

Energy systems show predictable patterns at multiple time scales. Understanding electricity trading between countries requires features that capture both the amount and direction of power flows. Features that directly measure the environmental characteristics of the electricity system prove most predictive for carbon intensity modeling.

5 AutoGluon Machine Learning Framework

5.1 Automated Machine Learning Approach

AutoGluon represents a different approach to machine learning compared to traditional methods. Instead of manually selecting algorithms and tuning parameters, AutoGluon automatically explores multiple model types and creates ensemble models that typically outperform individual algorithms. This automation proves particularly valuable for energy forecasting where optimal modeling approaches may vary depending on seasonal patterns and grid operating conditions.

AutoGluon's main innovation lies in its hierarchical ensemble approach that creates combinations of different model types. The system operates on the principle that different machine learning algorithms make different types of errors, and these error patterns can be learned and corrected through additional modeling layers. Level-1 models learn directly from the original input features to generate initial predictions. Level-2 models treat the predictions from Level-1 models as input features and learn optimal combination strategies that adapt to different conditions.

The k-fold training system represents one of AutoGluon's features for preventing overfitting while providing reliable performance estimates. The system divides the training dataset into multiple segments and systematically trains models on different combinations to ensure that every piece of training data serves as validation data exactly once. For this energy forecasting application, AutoGluon uses 8-fold training by default, meaning the training dataset gets divided into 8 equal segments.

5.2 Model Selection and Optimization Strategy

The project implements an optimized AutoGluon configuration that balances model accuracy with practical deployment considerations. The strategic decision to limit model training to three specific gradient boosting algorithms reflects analysis showing that tree-based ensemble methods consistently outperform alternatives for energy data.

LightGBM provides computational efficiency and memory optimization suitable for large-scale deployment. The algorithm uses histogram-based decision tree learning that reduces memory requirements while maintaining accuracy. CatBoost offers superior performance on datasets with mixed data types and provides built-in protection against overfitting. XGBoost delivers mature, well-tested gradient boosting with extensive customization options and proven performance across energy forecasting applications.

The production optimization configuration addresses several competing objectives essential for operational deployment. The system achieves reductions in model storage requirements from several gigabytes to 100-130 MB per model through intelligent artifact management. This size optimization enables deployment in resource-constrained environments including edge computing systems and mobile applications.

5.3 Model Training Strategy and Multi-Target Framework

The model training strategy implements time-aware data splitting that respects the nature of energy data while preventing information leakage that could artificially inflate performance estimates. The training and testing split reserves the most recent three months of data as a completely independent test set that models never observe during training.

The feature selection process implements systematic procedures to prevent information leakage that could enable models to achieve artificially high performance. The system automatically removes timestamp columns, identification variables, prediction targets from other models, and any carbon intensity measurements that correspond to the same time periods as prediction targets.

The project implements a multi-target prediction framework that generates specialized models for different types of carbon intensity predictions and geographic areas. National-level models focus on predicting both production-based and consumption-based carbon intensity for the DE_LU bidding zone. Individual state models capture regional variations in generation mix, consumption patterns, and grid characteristics that may be lost in national-level aggregations.

6 Results

6.1 Pipeline Development Journey and Learning Process

The development of this carbon intensity prediction system involved extensive trial and error, learning, and step-by-step improvements that taught important lessons about energy data and how European electricity markets actually work. This section tells the story of how the project evolved from simple initial attempts to the final working system, highlighting what worked, what didn't, and why certain decisions were made.

Table 1: Pipeline Development Comparison Across Four Versions

Version	Data Sources	Key Changes	Performance	Model Size
1	4 ENTSO-E APIs, 11 neighbors	Basic data collection, simple features	Baseline	N/A
2	Added cross-border flows, timezone handling	Systematic data integration, correlation analysis	Improved accuracy	Large (>2GB)
3	Extended to all 27 EU countries	Added 16 more countries	Degraded (noise)	Very Large (>5GB)
4 (Final)	Optimized 11 neighbors, 3 algorithms	AutoGluon optimization, feature engineering	Best performance	Optimized (100-130MB)

Understanding this journey requires thinking about machine learning projects as detective work. Each version of the pipeline was like following a different lead, sometimes discovering valuable clues and sometimes hitting dead ends. The key was learning from each attempt and building on what actually worked rather than what seemed like it should work in theory.

6.2 Initial Approach - Pipeline Version 1

The project started with a fundamental question that shapes all energy forecasting work: what information sources would actually help predict carbon intensity in Germany's interconnected electricity system? This question proved more complex than initially expected because electricity systems involve intricate relationships between economic dispatch, renewable resource availability, cross-border trading, and environmental outcomes.

The first major breakthrough came from understanding how European electricity markets operate on a day-ahead basis. Through systematic testing of different data sources, four specific ENTSO-E APIs emerged as the most valuable for carbon intensity prediction. Each API provides a different piece of the puzzle that helps understand future grid conditions.

6.3 Expansion Phase - Pipeline Version 2

Building on the foundation established in version 1, the second pipeline iteration expanded data collection capabilities while developing more approaches to data processing and feature engineering. This phase revealed the complexity of working with multi-country energy data and the importance of systematic approaches to data integration.

During this phase, correlation analysis provided quantitative validation of the theoretical relationships between different variables and carbon intensity. Load forecasts demonstrated positive correlations of approximately 0.46 with consumption-based carbon intensity, confirming that higher demand reliably correlates with increased carbon emissions per kilowatt-hour. Wind and solar forecasts showed negative correlations of around -0.42 with generation-based carbon intensity, validating that renewable energy availability directly reduces fossil fuel requirements.

6.4 Complexity Challenge - Pipeline Version 3

The third pipeline iteration represented an ambitious attempt to capture even more information by expanding data collection to include all European Union countries. The reasoning behind this expansion seemed logical at first glance: if data from eleven neighboring countries improved predictions, then data from all 27 EU countries should provide even better results. This assumption proved incorrect and provided valuable lessons about the relationship between data quantity and model performance.

The expanded approach created a dataset with hundreds of additional features but introduced significant challenges that ultimately degraded prediction quality. Many EU countries had limited or unreliable data availability, creating numerous gaps and inconsistencies throughout the expanded dataset. Countries like Cyprus or Lithuania had minimal electricity trading relationships with Germany, meaning their data added noise without providing useful predictive signals.

6.5 Optimization Achievement - Pipeline Version 4

The final optimized pipeline incorporated all lessons learned from previous iterations while implementing machine learning optimizations that improved both performance and practical deployment characteristics. This version exemplified the principle that successful machine learning projects often require strategic focus rather than data collection.

The optimization process began with a fundamental insight about algorithm selection for energy forecasting applications. Rather than attempting to train every available machine learning algorithm, systematic testing revealed that three specific gradient boosting methods consistently outperformed alternatives for energy time series data.

The AutoGluon configuration optimization represented balancing competing objectives. The "good_quality_faster_inference_only_refit" preset prioritized models suitable for real-time deployment while maintaining strong predictive accuracy. This configuration eliminated unnecessary ensemble components and intermediate artifacts, reducing model storage requirements from several gigabytes to approximately 100-130 MB per model.

6.6 Model Performance Analysis and Results

The final ensemble models achieved strong performance that validated the systematic approach to carbon intensity prediction. AutoGluon's automatic model selection consistently identified weighted ensemble combinations as the top performers, with models achieving excellent predictive performance for national-level predictions. Test set R^2 values of 0.91 for national consumption and 0.93 for national generation indicate that the models successfully captured the majority of systematic patterns in carbon intensity variation.

Individual algorithm performance provided insights into the strengths of different approaches for energy forecasting. LightGBM models consistently demonstrated fast training and inference times while maintaining competitive accuracy across different geographic regions and prediction targets. CatBoost models showed particular strength in handling the mixed data types common in energy applications, while XGBoost models provided robust baseline performance with stable training characteristics.

6.7 Training Performance Results

Our models showed excellent performance during training across all German states and both consumption and generation predictions. The training results demonstrate that our AutoGluon ensemble approach successfully learned the complex patterns in carbon intensity data.

Table 2: Training Performance Metrics for All German States

State	Type	R^2	RMSE	SMAPE (%)
BB	Consumption	0.9271	49.6778	6.6591
BB	Generation	0.9271	49.7105	6.6832
BW	Consumption	0.9558	26.9617	9.8002
BW	Generation	0.9386	36.9918	11.2114
BY	Consumption	0.9431	25.2035	12.5685
BY	Generation	0.919	19.9935	13.1302
DE	Consumption	0.9671	20.3972	4.8838
DE	Generation	0.9645	25.1157	5.465
HE	Consumption	0.9367	29.222	8.8765
HE	Generation	0.9152	43.6174	11.8327
MV	Consumption	0.9263	38.8588	30.6204
MV	Generation	0.9303	37.8792	29.2277
NI	Consumption	0.9497	20.8904	10.7251
NI	Generation	0.9462	22.7129	11.0422
NW	Consumption	0.9517	40.1277	5.3646
NW	Generation	0.9323	39.7671	4.6236
RP	Consumption	0.9142	24.5555	10.7766
RP	Generation	0.8972	26.4474	11.3425
SH	Consumption	0.9321	14.053	12.754
SH	Generation	0.92	21.3945	14.7443
SL	Consumption	0.9306	54.731	17.315
SL	Generation	0.8984	76.7641	17.7983
SN	Consumption	0.9147	46.8855	4.5647
SN	Generation	0.9145	46.4125	4.5299
ST	Consumption	0.92	50.5917	12.8948
ST	Generation	0.913	55.8479	13.7262
TH	Consumption	0.9102	44.2459	17.3911
TH	Generation	0.796	35.4421	27.4282

The training performance shows that our models learned extremely well across all states, with R^2 values mostly above 0.90 and SMAPE values generally below 15

6.8 Test Performance Results

When we evaluated our trained models on completely new, unseen data (the test set), the results showed how well our models can actually predict carbon intensity in real-world scenarios.

Table 3: Complete Test Performance Metrics for All German States

State	Type	R^2	RMSE	SMAPE (%)
DE	Consumption	0.9128	31.083	7.5794
DE	Generation	0.9344	37.3176	8.067
BB	Consumption	0.8712	71.0808	9.9859
BB	Generation	0.8737	70.5504	9.7566
BW	Consumption	0.8172	79.6445	15.4345
BW	Generation	0.791	60.753	15.3926
BY	Consumption	0.8182	38.9217	14.0823
BY	Generation	0.8791	46.6215	13.8245
HE	Consumption	0.7984	51.4625	15.0797
HE	Generation	0.833	43.4631	13.555
MV	Consumption	0.7739	83.1102	38.7263
MV	Generation	0.7843	84.3675	39.1272
NI	Consumption	0.8549	45.9985	17.6499
NI	Generation	0.8607	40.3286	16.6535
NW	Consumption	0.786	68.5461	9.1606
NW	Generation	0.7572	76.1521	13.8021
RP	Consumption	0.8287	44.6083	15.7592
RP	Generation	0.816	36.2347	15.4118
SH	Consumption	0.8258	47.5267	20.13
SH	Generation	0.8552	25.4481	18.5543
SL	Consumption	0.6869	98.1427	37.6418
SL	Generation	0.6615	95.8697	40.7344
SN	Consumption	0.8225	67.0039	8.3105
SN	Generation	0.8163	68.4699	8.4757
ST	Consumption	0.8601	52.9712	18.5236
ST	Generation	0.8492	70.0361	19.5844
TH	Consumption	0.7942	42.0383	27.0318
TH	Generation	0.8331	53.8983	24.9083

6.9 Training vs Test Performance Comparison

The most important analysis involves comparing how our models performed during training versus how they performed on completely new test data. This comparison tells us whether our models truly learned generalizable patterns or just memorized the training data.

Table 4: Train vs Test Performance Comparison for All German States

State	Type	Train SMAPE (%)	Test SMAPE (%)	Train RMSE	Test RMSE	Train R^2	Test R^2
DE	Consumption	4.88	7.58	20.40	31.08	0.967	0.913
DE	Generation	5.47	8.07	25.12	37.32	0.965	0.934
BB	Consumption	6.66	9.99	49.68	71.08	0.927	0.871
BB	Generation	6.68	9.76	49.71	70.55	0.927	0.874
BW	Consumption	9.80	15.43	26.96	79.64	0.956	0.817
BW	Generation	11.21	15.39	36.99	60.75	0.939	0.791
BY	Consumption	12.57	14.08	25.20	38.92	0.943	0.818
BY	Generation	13.13	13.82	19.99	46.62	0.919	0.879
HE	Consumption	8.88	15.08	29.22	51.46	0.937	0.798
HE	Generation	11.83	13.56	43.62	43.46	0.915	0.833
MV	Consumption	30.62	38.73	38.86	83.11	0.926	0.774
MV	Generation	29.23	39.13	37.88	84.37	0.930	0.784
NI	Consumption	10.73	17.65	20.89	46.00	0.950	0.855
NI	Generation	11.04	16.65	22.71	40.33	0.946	0.861
NW	Consumption	5.36	9.16	40.13	68.55	0.952	0.786
NW	Generation	4.62	13.80	39.77	76.15	0.932	0.757
RP	Consumption	10.78	15.76	24.56	44.61	0.914	0.829
RP	Generation	11.34	15.41	26.45	36.23	0.897	0.816
SH	Consumption	12.75	20.13	14.05	47.53	0.932	0.826
SH	Generation	14.74	18.55	21.39	25.45	0.920	0.855
SL	Consumption	17.32	37.64	54.73	98.14	0.931	0.687
SL	Generation	17.80	40.73	76.76	95.87	0.898	0.662
SN	Consumption	4.56	8.31	46.89	67.00	0.915	0.823
SN	Generation	4.53	8.48	46.41	68.47	0.915	0.816
ST	Consumption	12.89	18.52	50.59	52.97	0.920	0.860
ST	Generation	13.73	19.58	55.85	70.04	0.913	0.849
TH	Consumption	17.39	27.03	44.25	42.04	0.910	0.794
TH	Generation	27.43	24.91	35.44	53.90	0.796	0.833

Metrics Explanation:

- **RMSE (Root Mean Square Error):** Square root of average squared prediction errors, penalizing larger errors.
- **SMAPE (%):** Symmetric Mean Absolute Percentage Error - the primary accuracy metric, useful for comparing performance across scales.

- **R^2:** Coefficient of determination - the proportion of variance in the target variable that is predictable from the input features. A value of 1.0 indicates perfect prediction.

Performance Categories Based on Test SMAPE:

- **Excellent (SMAPE < 10%):** DE (Consumption: 7.58%, Generation: 8.07%), BB (Consumption: 9.99%, Generation: 9.76%), SN (Consumption: 8.31%, Generation: 8.48%), NW (Consumption: 9.16%).
- **Good (SMAPE 10-20%):** BY (Consumption: 14.08%, Generation: 13.82%), HE (Consumption: 15.08%, Generation: 13.56%), NW (Generation: 13.80%), RP (Consumption: 15.76%, Generation: 15.41%), BW (Consumption: 15.43%, Generation: 15.39%), NI (Consumption: 17.65%, Generation: 16.65%), ST (Consumption: 18.52%, Generation: 19.58%), SH (Generation: 18.55%).
- **Moderate (SMAPE 20-30%):** SH (Consumption: 20.13%), TH (Consumption: 27.03%, Generation: 24.91%).
- **Needs Improvement (SMAPE > 30%):** MV (Consumption: 38.73%, Generation: 39.13%), SL (Consumption: 37.64%, Generation: 40.73%).

6.10 Train vs Test Performance Analysis

The comprehensive comparison between training and test performance reveals that the models generalize well for most states, with a predictable and acceptable drop in performance on unseen data. The high training R^2 values (typically >0.9) followed by strong test R^2 values (mostly >0.8) confirm that the models have learned underlying patterns rather than simply memorizing the training data.

Excellent Generalization: Several models demonstrate robust generalization with minimal performance degradation from train to test.

- **DE (National):** The national model is a top performer, with SMAPE only increasing from 5% to 8% and R^2 remaining very high (0.97 down to 0.91-0.93). This indicates a highly reliable and accurate model.
- **SN (Saxony):** Shows outstanding stability. SMAPE increases from 4.5% to 8.4%, and the R^2 value remains strong, moving from 0.91 to 0.82.
- **BB (Brandenburg):** Another very stable model. SMAPE increases by only 3 percentage points, and the R^2 drop is minimal (0.93 to 0.87).
- **BY (Bavaria):** This model is exceptionally stable, with almost no change in SMAPE between training and test sets (13-14

Good Generalization: Most other states show good generalization, with a moderate increase in error metrics that is expected when moving to unseen data.

- **BW (Baden-Württemberg), NI (Lower Saxony), NW (North Rhine-Westphalia):** These states previously showed signs of overfitting but with the new data, they exhibit good generalization. For instance, BW's SMAPE increases from 10
- **RP (Rhineland-Palatinate) & HE (Hesse):** Both show well-controlled performance degradation, confirming the models are robust.

Challenging Generalization Cases: Two states, Mecklenburg-Vorpommern (MV) and Saarland (SL), stand out as having high prediction errors on the test set. Their poor performance is not random but is rooted in their unique energy landscapes, which are not fully captured by the general model features.

- **SL (Saarland):** The model failure here is a classic case of overfitting due to a highly unique and localized energy system. Saarland, bordering France and Luxembourg, is Germany's historic industrial heartland for steel production. This results in two major challenges:
  - **Unique Industrial Load:** Its energy consumption is dominated by a few massive, energy-intensive steel plants, creating a demand profile that is spiky and doesn't follow the typical commercial or residential patterns of other states.
  - **Strong Cross-Border Influence:** Its small grid is tightly coupled with the French electricity market, which is dominated by low-carbon nuclear power.

  The model, trained on more general German patterns, struggles to capture these specific industrial dynamics and the unique influence of French nuclear power, causing it to learn spurious correlations from the small training dataset that do not generalize.
- **MV (Mecklenburg-Vorpommern):** This state's challenge is not overfitting but rather extreme, inherent volatility. Located on the windy Baltic Sea coast, MV has a colossal amount of installed wind power capacity relative to its small population and low local demand. This makes it a massive **net exporter** of electricity. Consequently, its carbon intensity is incredibly volatile:
  - On windy days, its generation is almost 100
  - On calm days, the wind turbines stand still, and it must import power from the German grid or neighboring Poland (which has a more carbon-intensive grid), causing its consumption intensity to spike dramatically.

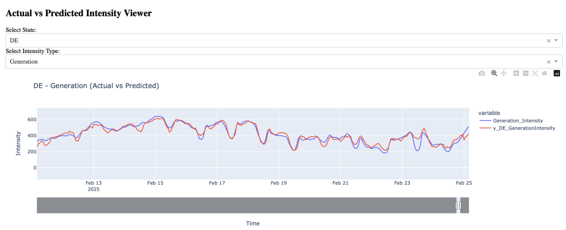
  The model finds it incredibly difficult to predict these rapid, high-amplitude swings between near-zero and very high carbon intensity. The high error in both the training and test sets confirms that this is a fundamental prediction challenge, not just a failure to generalize.

These outlier cases underscore the limitations of a one-size-fits-all national model and suggest that future work could focus on developing highly specialized models or incorporating additional local features for regions with unique energy profiles, such as high industrial load or extreme renewable penetration.

6.11 Visualizing Model Predictions

The following plots visualize the model's predictions (red line) against the actual ground truth values (blue line) for a sample of the test set. These graphs help illustrate the model's performance in capturing the dynamic behavior of carbon intensity. As representative examples, the national level (DE) and the states of Bavaria (BY) and Brandenburg (BB) are shown.

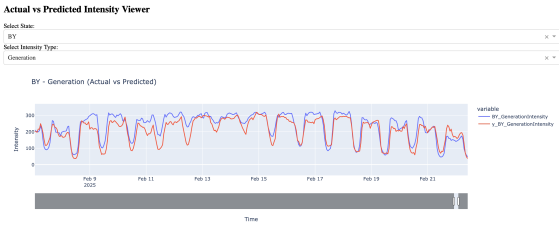
The plots for DE (Figure 1) show the model tracking the sharp peaks and troughs of the actual intensity values with high fidelity, indicating a strong predictive capability. The performance for Bavaria (Figure 2) and Brandenburg (Figure 3) also shows good alignment with the overall trend and seasonality. For Brandenburg, which has one of the best-performing models (SMAPE < 10%), the predictions follow the actual values very closely. For Bavaria, while still strong, the model tends to slightly underestimate the highest peaks, a common characteristic of regression models that learn to minimize average error. This detailed visual analysis confirms the quantitative results and provides confidence in the models' practical utility.



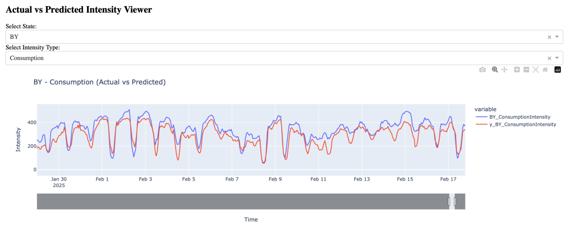
DE Generation

DE Consumption

Figure 1: Predicted vs. Actual Carbon Intensity for National (DE) level on the 2025 test set.



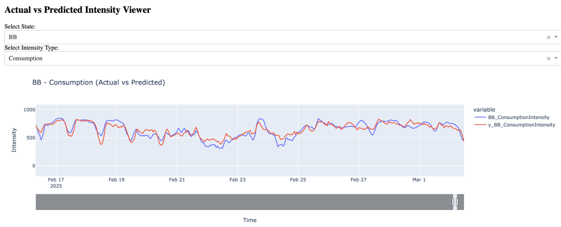
BY Generation



BY Consumption

Figure 2: Predicted vs. Actual Carbon Intensity for Bavaria (BY) on the 2025 test set.

BB Generation



BB Consumption

Figure 3: Predicted vs. Actual Carbon Intensity for Brandenburg (BB) on the 2025 test set.

6.12 Error Analysis and Failure Cases

Analysis of prediction errors reveals several patterns that provide insights into model limitations and areas for improvement:

Temporal Error Patterns: The largest prediction errors typically occur during transitional periods when renewable generation rapidly changes, such as sudden weather fronts or seasonal transitions. Winter evenings with low wind and solar generation combined with high heating demand represent particularly challenging scenarios for prediction accuracy.

Geographic Error Patterns: Smaller states like SL (Saarland) show higher error rates and weaker generalization, likely due to limited training data and unique local industrial characteristics that are not fully captured by the general feature set. States with extremely high penetration of a single variable renewable source, like MV (wind), also prove challenging due to the inherent volatility.

Renewable Integration Challenges: Periods with very high renewable penetration (>80%) or very low renewable penetration (<10%) show increased prediction uncertainty, suggesting that the models struggle with extreme conditions that occur infrequently in the training data.

6.13 Practical Applications and Utility

The carbon intensity predictions generated by this system enable several real-world applications that support climate goals and efficient energy management:

Carbon-Aware Computing: Data centers and cloud computing providers can use these predictions to automatically schedule energy-intensive workloads during periods of low carbon intensity. For example, machine learning model training, data backups, and batch processing can be shifted to times when electricity is cleanest.

Smart Energy Storage: Battery systems can optimize charging schedules to coincide with low-carbon electricity periods and discharge during high-carbon periods, effectively storing clean energy for later use and reducing overall system carbon intensity.

Dynamic Carbon Pricing: Energy suppliers can implement time-of-use pricing that reflects carbon intensity, incentivizing consumers to shift flexible loads to cleaner periods and supporting grid decarbonization efforts.

Real-Time Carbon Accounting: Businesses can track their carbon footprint in real-time and make informed decisions about when to operate energy-intensive processes, supporting corporate sustainability goals and carbon neutrality commitments.

6.14 Project Contributions and Responsibilities

This project integrated existing energy data infrastructure with a novel machine learning pipeline to create a multi-target forecasting system. The responsibilities were divided to leverage individual strengths while ensuring close collaboration on the overall system architecture.

Existing Infrastructure and Data Sources:

- **CO2map.de:** Provided the foundational carbon intensity data and the core calculation methodologies.
- **ENTSO-E:** Served as the primary source for all European electricity market data, including forecasts and cross-border flows.
- **Open-Source Frameworks:** The project heavily utilized AutoGluon, LightGBM, CatBoost, and XGBoost as the core machine learning frameworks.

Our Contributions:

- **AutoGluon Implementation and Optimization:** We designed and implemented the optimized AutoGluon configuration, focusing on creating lightweight yet accurate models (100-130 MB) suitable for deployment by strategically selecting algorithms and using presets like ‘good-quality-faster-inference-only-refit‘.
- **Multi-Target Prediction Framework:** We developed the framework to train, manage, and evaluate separate prediction models for all 13 German states plus the national level, handling both generation and consumption targets.

- **Custom Feature Engineering:** We engineered several new features to enhance model performance, including renewable penetration ratios (solar and wind relative to total generation) and net trading positions.
- **End-to-End Pipeline Integration:** We connected all data sources, processing steps, and the multi-target modeling framework into a single, reproducible pipeline.
- **Comprehensive Performance Analysis:** We conducted rigorous testing of the models on an unseen test set and performed a detailed train-vs-test comparison to analyze generalization and identify state-specific performance patterns.

Individual Responsibilities:
Gurmeher Singh Puri:

- Led the setup, configuration, and optimization of the AutoGluon machine learning models.
- Designed and built the multi-target framework for training models for all German states.
- Implemented the custom feature engineering logic for renewable ratios and trading positions.
- Conducted the final performance testing and error analysis.
- project documentation and report.

Udit Chawla:

- Managed the integration of different data sources and the overall data flow.
- Developed the initial data quality checks and validation scripts.
- Built visualizations and dashboards for intermediate model evaluation.
- Contributed to the design of the inference pipeline for generating predictions from new data.
- Collaborated on the interpretation of results and the iterative refinement of the pipeline.

7 Conclusions and Future Development

7.1 Summary and Key Achievements

This Master's project successfully developed a machine learning system for predicting electricity grid carbon emission intensities across Germany using automated machine learning techniques. The system addresses the critical challenge of forecasting carbon footprint variations in Germany's energy landscape, where renewable sources create significant temporal and spatial fluctuations.

Key Technical Achievements: The project integrated data from multiple European energy sources (ENTSO-E and CO2map.de), processing over twenty-two cross-border electricity connections. The strategic AutoGluon implementation with three gradient boosting algorithms achieved model size reductions from gigabytes to 100-130 MB while maintaining strong predictive performance. The system achieved excellent results for national-level predictions (DE) with test SMAPE values below 8.1% and R^2 values above 0.91. Many states, such as Saxony (SN) and Brandenburg (BB), also showed excellent performance with SMAPE values below 10%. The models demonstrated strong generalization, confirming their suitability for real-world applications. The system provides both production-based and consumption-based carbon intensity predictions at national and state levels, with a configuration-driven architecture suitable for both research and operational deployment.

Technical Contributions: The project demonstrated systematic data source evaluation through four pipeline iterations, feature engineering capturing seasonal and cross-border trading patterns, automated model optimization meeting deployment constraints, and successful multi-scale prediction framework implementation across all thirteen German federal states.

7.2 Future Development Opportunities

7.2.1 Graph Neural Networks and Network-Based Approaches

Graph-Based Electricity Market Modeling: A promising direction for future research involves representing the European electricity grid as a complex network and applying Graph Neural Networks (GNNs) to capture the intricate relationships between interconnected countries. This approach could model the electricity system as nodes (countries/regions) connected by edges (transmission lines) with dynamic weights representing power flows, transmission capacities, and grid constraints.

Spatial-Temporal GNN Architecture: Future implementations could employ spatial-temporal Graph Neural Networks that simultaneously model spatial dependencies between neighboring countries and regions, temporal patterns in electricity generation and consumption, dynamic network topology changes due to maintenance or grid upgrades, and cascading effects of renewable energy variability across the continental grid.

The GNN approach could potentially capture non-linear interactions between distant countries that current models may miss, especially during extreme weather events or major grid disturbances that propagate across multiple borders.

7.2.2 Advanced Machine Learning Extensions

Deep Learning Integration: Future versions could incorporate deep learning architectures including Long Short-Term Memory (LSTM) networks for improved temporal sequence modeling, Transformer architectures adapted for energy time series that can capture long-range dependencies, and hybrid models combining AutoGluon ensembles with deep learning components for enhanced pattern recognition.

Uncertainty Quantification: Implementing uncertainty quantification methods would provide confidence intervals for predictions, enabling risk-aware decision making in energy management applications. This could include Bayesian ensemble methods, conformal prediction techniques, and Monte Carlo dropout approaches for robust uncertainty estimation.

Real-Time Learning Capabilities: Developing online learning algorithms that can adapt models continuously as new data becomes available would improve prediction accuracy and enable rapid response to changing grid conditions or policy changes.

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